



Paper 1 (CORE and APPLICATIONS)
MCQ

1	A	6	B	11	B	16	A	21	C	26	D	31	B	36	D
2	B	7	A	12	B	17	A	22	B	27	A	32	A	37	C
3	C	8	B	13	D	18	D	23	D	28	C	33	C	38	A
4	D	9	B	14	C	19	B	24	A	29	C	34	A	39	D
5	B	10	D	15	B	20	B	25	B	30	B	35	D	40	D

Paper 2 (CORE)
Section A: Structured

QUESTION 1

(a)(i) Define the term glycoprotein. [1]

- Protein / polypeptide;
 - with oligosaccharides / carbohydrates;
 - covalently attached;
- (1m max)

(ii) Suggest a role of PrP in the membranes of nerve cells. [1]

- Cell signaling / cell membrane receptors receiving signals from ligands;; OR Cell adhesion;;
- (1m max)

(b)(i) With reference to **Fig. 1.1 & Fig. 1.2**, relate how the differences in the structure of PrP and PrP^{SC} bring about the differing effects of enzyme action. [2]

PrP^{SC} has

- higher proportions of β -sheets than PrP;
- i.e. 4 in PrP^{SC} as compared to 1 in PrP;

OR

- lower proportion of α -helices than PrP;
- i.e. 2 in PrP^{SC} as compared to 3 in PrP;

(1m max)

- There are differences in the folding / bending / twisting; of the PrP and PrP^{SC} proteins
 - which give rise to different 3-D conformations;.
 - Protein is no longer complementary to the active site of enzyme;
 - Protein unable to form enzyme-substrate complex / bind to the active site of enzyme;
- (1m max)

(c) With reference to **Fig 1.3**,

(i) identify the evidence that suggests the experimenters tested only one sample in each of the controls. [1]

- Absence of error bars;
 - No variation observed;
 - i.e. % PrP^{SC} after 48 hrs remained at 100%; for both mouse and human
 - at 0:1 ratio;
- (1m max)

(ii) suggest **two** conclusions about the effect of β -breaker on human and mouse PrP^{SC}. [2]

Any two of the following:

β -breaker decreases % PrP^{SC}

- As ratio of β -breaker to PrP^{SC} increases, % PrP^{SC} decreases in mice and human;
 - from 100% at ratio 0:1 to 15% at ratio 1000:1 for mice;
 - and from 100% at ratio 0:1 to 37.5% at ratio 100:1 for human;
- (1m max)



Comparison between mice and human.

- % PrP^{SC} present after 48 hours is more in human than mice at all ratios;
 - There is a bigger % PrP^{SC} decrease in mice than humans for each ratio;
 - thus, the β -breaker works better / greater effect in mice;
- (1m max)

Mice

- In mice there is less % PrP^{SC} decrease as β -breaker concentration increases;
- There is a drop of % PrP^{SC} from 52.5% at ratio 10:1 to 19.5% at ratio 100:1 but a drop from 19.5% at ratio 100:1 to 15% at ratio 1000:1 OR There is a drop from 100% at ratio 0:1 to 50% at ratio 1:1 but a drop from 19.5% at ratio 100:1 to 15% at ratio 1000:1;
- which indicates that β -breaker gets less effective at higher concentrations;
- β -breaker is less effective for mice at 10:1 ratio as % PrP^{SC} present showed an increase from 50% to 52.5%;

(1m max)

Human

- Higher concentration of β -breaker may not be more effective with decreasing % PrP^{SC};
- Best effect in humans is 100:1 when the % PrP^{SC} decrease to 37.5%;
- Larger variation in digestion of human PrP^{SC};
- There is a drop of % PrP^{SC} from 100% at ratio 1:1 to 37.5% at ratio 100:1 but a increase from 37.5% at ratio 100:1 to 50% at ratio 1000:1;

(1m max)

(d) With reference to **Table 1.4**,

(i) Explain how two conclusions can be derived from the trends observed.

[3]

- The lower the concentration of PrP^{SC} the greater the delay in onset of symptoms;
- As concentration of PrP^{SC} increases from 1×10^{-2} to 1×10^{-4} , the time to onset of symptoms increases from 129 days to 173 days;
- PrP^{SC} is an infectious agent;

(1.5m max)

- The presence of β -breakers increases the delay in onset of symptoms;
- for all concentrations of PrP^{SC} from 143 days to 185 days;
- The β -breaker is effective in delaying the onset of the symptoms;
- Once infected by PrP^{SC}, there is still no cure;

(1.5m max)

(ii) suggest **one** modification to the treatments in the second study that may delay the onset of symptoms further.

[1]

Any one of the following:

- Use of higher concentration of β -breaker;;
- Incubate PrP^{SC} with β -breaker (before injection)/incubate longer/different length of incubation;;
- Reduce the PrP^{SC} concentration;;
- Use optimum temperature for β -breaker to work optimally;;
- Use optimum pH for β -breaker to work optimally;;

(1m max)

[Total: 11 marks]

QUESTION 2

(a) Identify molecules **A** and **H**.

[2]

A: Citrate ;;

H: Oxaloacetate ;;

(b) Indicate the number of carbon atoms represented by **Y** and **Z**.

[1]

Y: 5 ;

Z: 6 ;

(c) With reference to **Fig 2.1**, give one reason why the Krebs cycle is called a cycle.

[1]



The starting substrate H; is regenerated at step 8 which is the last step of the cycle;

(d) For every molecule of glucose that is completely oxidized, explain how many pairs of electrons are stripped from the respiratory substrates in the Krebs cycle. [2]

- For every glucose molecule, there are 2 turns of the Krebs cycle; which generates 2 FADH_2 and 6 NADH ;
- Since each co-enzyme carries 1 pair of electrons ; There will be 8 pairs of electrons being stripped ;

(e) State the identities of **P** and **Q**. [1]

P: NADH ;

Q: NAD ;

(f) Describe the significance of **P** and **Q** in linking the Krebs cycle to the formation of ATP in the cristae. [2]

- **Q** which is NAD receives electrons stripped from respiratory intermediates in the Krebs cycle which completes the oxidation of glucose ; and is reduced to **P** which carries the high-energy electrons to the electron transport chain (ETC) at the inner mitochondrial matrix ;
- Energy released during the passage of electrons down the ETC ; is coupled to the formation of a proton gradient ; which will be harnessed by ATPase to power chemiosmosis / is coupled to chemiosmotic synthesis of ATP ;

(g) Explain the significance of **S** to the muscle cell. [2]

- In the event of high ATP demand e.g. a sprint, the rate of oxygen taken in / supply cannot meet demand/ absence of oxygen in the cell;
- Hence the muscle cell goes into **S** which is anaerobic respiration / lactic acid fermentation ; to regenerate NAD when pyruvate is reduced to lactate ; for production of ATP by SLP in glycolysis ;
- Lactic acid will be slowly converted to pyruvate in the liver when oxygen is available later on / oxygen debt re-paid ; allowing the energy trapped in lactic acid to re-enter metabolism/ Krebs cycle / used later on;

[Total: 11 marks]

QUESTION 3

(a)(i) State a possible identity for this gene regulatory protein. [1]

- (General) transcription factor;; OR Activator;;

(ii) Describe how this gene regulatory protein identified in **(a)(i)** could result in higher transcriptional efficiencies. [2]

- Binding of this (general) transcription factor to the core promoter / promoter-proximal elements;
- OR
- Binding of activator to enhancer;
 - recruits other DNA-binding proteins / transcription factors / activators / co-activators;
 - to form a stable transcription initiation complex; (resulting in an increase in transcription levels)

(b)(i) With reference to **Fig. 3.1** and **Fig. 3.2**, compare the effects of the different deletions between the purified components and the nuclear extract. [2]

- Deletion up to 11 bp upstream of transcription start site / -11 deletion template inactivates transcription in both systems;
- Deletions up to 61 bp upstream of transcription start site / -61, -81, -127 deletion templates have no effect on transcription level / results in the same transcription level as compared to the undeleted template in both systems;
- Deletion up to 50 bp upstream of transcription start site / -50 deletion template results in a lower transcription level in the nuclear extract; which is comparable to the transcription level for deletions up to 50 bp upstream of transcription start site / -50 deletion template in the purified system;

(ii) Account for the absence of a band for the -11 deletion template in both the purified system and the nuclear extract as shown in **Fig. 3.2**. [2]



- Deletions up to 11 bp upstream of transcription start site / -11 deletion template would imply that the TATA box (which resides ~30 bp upstream of transcription start site) is deleted;
- meaning that general transcription factor cannot bind; resulting in no recruitment of RNA polymerase to the core promoter; and thus no transcription / no mRNA transcripts formed; (resulting in no bands formed for both systems.)

(iii) Deduce the binding site of the gene regulatory protein. [1]

- The binding site for the gene regulatory protein must be between 50 – 61 bp / -61 and -50 nucleotides upstream of transcription start site;;

(iv) Give a reason for your answer to **(b)(iii)**. [1]

- The absence of binding site for the gene regulatory protein in the -50 deletion template means it cannot bind and thus cannot increase the transcription level in the nuclear extract;
- unlike in the -61, -81 and -127 deletion templates where its binding site is present and thus it can bind to increase the transcription levels in the nuclear extract;

(c)(i) State a possible identity for SAGA. [1]

- Co-activator;;

(ii) Suggest how SAGA can affect gene expression. [2]

- SAGA has histone acetyltransferase (HAT) which catalyzes the transfer of acetyl groups (from acetyl CoA) to positively-charged lysines in the histone (N-terminal) tails; resulting in lysines becoming uncharged;
- This reduces affinity of histone tails for DNA; which results in the loosening of histone complex from DNA; thereby exposing DNA regions to transcription machinery; and increasing the rate of transcription;

OR

- SAGA also contains subunits that interact with general transcription factors; which facilitates recruitment of RNA polymerase II to bind to core promoter; and also helps to stabilize the transcription initiation complex;
 - This results in higher rate of transcription of the entire coding region of DNA into mRNA;
- (2m max)

[Total: 12 marks]

QUESTION 4

(a) Define the term inducible enzymes, with respect to β -galactosidase. [1]

- Synthesis of β -galactosidase can be stimulated when lactose is available;
- β -galactosidase is not normally produced in cells;

(b) Explain how the regulation at the operator and CAP-binding site differs from each other. [2]

- Repressor binds to operator; while activated CAP binds to CAP-binding site;
- Binding of repressor to operator prevents transcription while binding of activated CAP increases the rate of transcription;
- The operator is involved in negative control; while the CAP-binding site is involved in positive control;

(c) Name the process of this gene transfer into the bacterial chromosomes by bacteriophages. [1]

- Specialized; transduction;

(d) Suggest **two** advantages of the process of gene transfer named in **(c)** over transformation in bacteria. [2]

- The gene can be integrated into the chromosome of bacteria in transduction; but it may not be integrated in transformation;
- The gene is protected from nuclease digestion by the capsid of bacteriophage during the transfer by transduction; but the unprotected and naked DNA during the transfer in transformation can be digested;
- The rate of successful gene transfer is higher in transduction; than in transformation;

(e) Explain why the gene expression is measured as “ β -galactosidase activity in the presence of inducer divided by β -galactosidase activity in the absence of inducer”. [1]



- To measure the degree of repression/down regulation/negative regulation by the presence of the different sequences;
- The greater the ratio, the higher the repression/down regulation/negative regulation;

(f) With reference to **Fig. 4.1**, discuss the roles played by S and T in *lac Z* gene expression. [2]

- In the presence of R, S and T sequences, the down regulatory effect/ability to prevent transcription of *lac Z* gene is at maximum, with a ratio of 1300 as seen in construct A;
- T is essential to the down regulatory effect/to prevent transcription of *lac Z* gene; as seen in gene constructs D and E where the ratio is 1 when T sequence is replaced;
- S contributes partially to the down regulatory effect/to prevent transcription of *lac Z* gene; as seen in gene construct B where the ratio reduce from 1300 to 450 when S sequence is replaced;

(g) Suggest how could changes in DNA sequence within the *lac Z* gene, as indicated by the crosses (X) in **Fig. 4.1**, could still result in the production of functional β -galactosidase. [2]

- In silent mutation; due to degeneracy of the genetic code; the same amino acid can be coded by more than one codon; so changes at the third nucleotide of a codon; will not result in a change in amino acid sequence of the enzyme;
- In neutral mutation; a change in DNA sequence could result in different amino acids, but with similar chemical/physical properties being translated; so the overall tertiary structure/3D configuration of the enzyme remains unchanged; / the change in DNA sequence results in different amino acids, which are non-essential to the function of the enzyme; so the overall tertiary structure/3D configuration of the enzyme remains unchanged;

[Total: 11 marks]

QUESTION 5

(a) Using evidence from **Fig. 5.1**, explain why it is unlikely that the disease is a sex-linked recessive trait. [2]

- If sex-linked recessive, affected mother I-2 who would be homozygous recessive; should produce sons that are all affected; as they inherit the X-linked disease allele from I-2; but her two sons II-1 and II-2 are unaffected; OR
- Affected female V-6 who would be homozygous recessive; should have an affected father IV-5; from whom she inherited one of her two copies of the X-linked disease allele; but her father IV-5 is unaffected; OR
- Unaffected male IV-5 who would be hemizygous dominant; should have unaffected daughters; as they receive the X-linked dominant normal allele from him; but he has one affected daughter V-6;

(b) Individual **V-3** married individual **V-4** and the couple plans to have a child. Calculate the probability that their first child will be affected with the disease, given that pigment levels are controlled by a single autosomal gene which has two alleles. [1]

- 1/16;;

(c) Explain why affected individuals in Generation I can produce normal children. [2]

- Affected individuals are homozygous recessive; and married normal individuals who are either homozygous dominant; or heterozygous carriers;
- thus at least one copy of the normal dominant allele is transmitted to their children; who will thus be phenotypically normal as the dominant normal allele from the unaffected parent masks the effect of the recessive disease allele (from the affected parent);

(d) Draw a genetic diagram of the F_1 cross to show the predicted results, indicating clearly the expected proportions of the different phenotypic classes in the F_2 offspring. [4]

Let: Pr be the dominant allele for purple eyes
 Pr+ be the recessive allele for wild-type eyes ;
 e+ be the dominant allele for wild-type body
 e be the recessive allele for ebony body ;

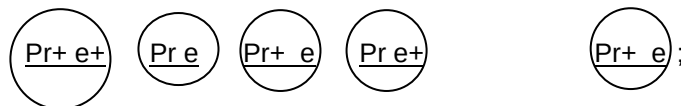
F_1 Phenotypes Purple eyes, Wild-type eyes,
 wild-type bodies ebony bodies
 (female) x (male)

F_1 Genotypes Pr+ e+ Pr+ e



Pr e ; Pr+ e ;

F1 Gametes



Random fertilisation

		Female gametes			
Male gametes		$\frac{\text{Pr+ e+}}{\text{Pr+ e}}$	$\frac{\text{Pr+ e}}{\text{Pr e}}$	$\frac{\text{Pr+ e}}{\text{Pr+ e}}$	$\frac{\text{Pr e+}}{\text{Pr+ e}}$;

F2 genotypic ratio 44.5 $\frac{\text{Pr+ e+}}{\text{Pr+ e}}$: 44.5 $\frac{\text{Pr+ e}}{\text{Pr e}}$: 5.5 $\frac{\text{Pr+ e}}{\text{Pr+ e}}$: 5.5 $\frac{\text{Pr e+}}{\text{Pr+ e}}$;

F2 phenotypic ratio

44.5 Wild-type eyes, wild-type body	44.5 Purple eyes, ebony body	5.5 Wild-type eyes, ebony body	5.5 Purple eyes, wild-type body
parentals		recombinants	

(e) Explain how the proportions of the different F₂ phenotypic classes obtained in (d) are derived. [2]

- The distance between the eye colour and body colour genes is 11 cM, implying that the COV / recombination frequency of the two genes is ~11%;
- Since the percentage of crossing-over is ~11%, the probability of obtaining each of the two possible recombinant gametes in the F1 females is 11/2 = 5.5%; which is equivalent to the proportion of each of the two classes of recombinant F2 offspring;
- The rest of the offspring falls into two classes of parentals, the proportion of each being (100-11)/2 = 44.5%;

(f) Assuming that double-crossing does occur between the two genes, suggest why the recombination frequency between the two loci would deviate from the expected percentage. [1]

- A double cross-over returns / restores the grouping of the linked alleles to the parental combination; thus the recombination frequency between the linked loci will be lower than expected;

[Total: 12 marks]

QUESTION 6

(a) Describe the events which occur when an action potential reaches the synaptic knob prior to the release of neurotransmitters into the synaptic cleft. [2]

- Depolarisation of the presynaptic knob; occurs to cause the voltage-gated Ca²⁺ channels to open; allowing an influx of Ca²⁺ ions; into the presynaptic knob.
 - Ca²⁺ is necessary in activating neurofibrils to move synaptic vesicles containing neurotransmitter molecule; to fuse to presynaptic membrane; releasing the neurotransmitter into the synaptic cleft.
 - SNARE proteins also mediate the fusion of those synaptic vesicles to the presynaptic membrane;
- (2m max)

(b) With reference to Fig. 6.1,

(i) deduce the relationship between EJP amplitude and the average number of synaptic vesicles per synapse in the SHi mutants at both temperatures. [2]

- At 19°C, as the average number of vesicles per synapse increases from 5 to 50, there is a linear increase of EJP amplitude from 6mV to 40mV;;
- At 29°C, as the average number of vesicles per synapse increases from 5 to 50, there is a linear increase of EJP amplitude from 12mV to 70mV;;

(ii) explain your answer in (b)(i). [2]

- In the SHi mutants, the more the synaptic vesicles present in the presynaptic knob, the more the amount of neurotransmitter that will be released into the synaptic cleft; and diffuse across the synaptic cleft; to bind to its specific postsynaptic membrane receptors;.



- More chemically-gated Na^+ channels open; leading to a greater Na^+ influx;
- This results in more depolarisation of postsynaptic membrane; and hence, higher EJP values recorded;

(c) In addition to myelination, suggest **one** way in which the structure of the postsynaptic neuron is adapted to its function. [1]

Many mitochondria; in cell body and in axon	Produce ATP; for sodium/potassium pumps – needed to maintain resting potential.
The axon has a wide diameter;	Increase the speed of conduction; of a nerve impulse.
The branched terminal ends; of axon	Increase surface area; so that the axon can form may synapse with postsynaptic neuron.
The membranes have both sodium and potassium channels; at the nodes of Ranvier.	This allows the regeneration and transmission of an action potential;.

(d) Explain how the same glutamate can have different effects on different cells. [2]

- Upon signal reception; the same glutamate binds to different receptors; on the cell membrane of different cells in the midgut and hindgut;
- Different ligand-receptor complexes; are formed.
- Both the midgut and hindgut have different signal transduction pathways; which cause different cellular responses; in the midgut and hindgut respectively.

(e) Compare between neural signalling and endocrine signalling. [2]

Any one of the following similarities:

- Both types of signalling can occur over long distances;;
- Both forms of signalling are part of the cell-to-cell communication system;; in the body.
- Both produce specific cellular responses;;

(1m max)

Any one of the following differences:

- Neuronal signalling is very fast; hormonal signaling is slower;
- Nerves communicate directly with one or a few neurons but hormones communicate with all the diverse and widely dispersed target cells in the body.
- Response in neural signalling is often very localized; compared to very widespread response in endocrine signalling;
- Mode of transmission in neural signalling is via the nerve cells / nervous system; while mode of transmission in endocrine signalling is via the bloodstream / circulatory system;

(1m max)

[Total: 11 marks]

QUESTION 7

(a)(i) Describe **two** important features of the binomial nomenclature. [2]

- A naming system;
- Each species of organism is assigned a unique; two-part; name
- The first part of a scientific name is designated the genus; and the second part is the species / specific epithet;

(ii) Scientific names of organisms are often difficult to spell, pronounce and remember. Despite this, scientific names are frequently used rather than common names. Suggest two advantages for the use of scientific names rather than common names. [2]

Any two advantages:

- Each species has a unique scientific name which is used internationally / universally;; so each organism's name is very specific; and cannot be misinterpreted;
- Some organisms do not have a common name;; so can only be referred to by their scientific name;
- As scientific names are in Greek/Latin;; their meaning is not going to change over time, as will common names in other languages;
- Common names can cause confusion; as several species have the same common name that may vary from region to region, or include unrelated species;



(b) Describe what is meant by the term “cladogram”.

[2]

A cladogram:

- is a phylogenetic tree;
- is based on shared derived characters;
- includes an ancestor and all its descendents;
- is a form of monophyletic grouping / clade;
- shows evolutionary relationship;

R: it is a graph / diagram.

(c) With reference to **Fig. 7.1** and **Fig. 7.2**,

(i) explain the role of shared derived traits in generating a cladogram for evolution of birds.

[2]

- Shared derived characters are homologous traits;
- that originate in a recent;
- common ancestor;
- present in its descendants;
- aids in tracing common ancestry;
- determines branching points in the cladogram;
- shows divergence;

(1.5m max)

- presence of flapping wings / presence of flight stroke / presence of half-moon shaped wrist bone from *Velociraptor* to *Columba* but not observed in *Coelophysis*;

(0.5m max)

(ii) With reference to **Fig. 7.1**, give a reason why *Archaeopteryx* appears to be most closely related to dinosaurs.

[1]

Archaeopteryx has similar morphological features as dinosaurs that include (any one of the following):

- presence of half-moon shaped wrist bone;;
- fusion of finger bones I to III;;
- same arrangement of finger bones;;
- no flapping of wings;;

(iii) compare the flight capabilities of *Velociraptor* and *Columba*.

[2]

Any one of the following differences:

- *Velociraptor* had no flight capability; while *Columba* has flight capability;
- bones fuse / less in *Columba*; and separate / more in *Velociraptor*;

(1m max)

Any one of the following similarities: Both have

- flapping of wings;;
- three fingers;;
- half-moon-shaped wrist bone;;
- finger bone / digit II as longest;;
- hollow bones;;

(1m max)

(d) Outline what new evidence would be needed to come to this conclusion.

[2]

New evidence would include:

- Other paleontological evidence / fossils; e.g. other skeletal structures;
- Biogeographical evidence of where bird fossils are found;
- Position of fossils in the rocks; comparing the positions of the fossil birds relative to one another;
- Molecular evidence; e.g. nucleic acids / amino acids homologies; to estimated from differences due to mutations in living birds;

Purpose of using new evidence:

- To compare; and deduce evolutionary relationships;
- To observe if the different bird lineages did diverge from a common ancestor;;
- To observe divergent evolution;

[Total: 12 marks]



QUESTION 8

(a) Explain how the process of DNA replication maintains the integrity of genetic information and at the same time allow for possible occurrences of mutation. [10]

- A1. DNA replication ensures the synthesis of an exact copy of DNA;
- A2. based on a semi-conservative model of replication;
- A3. where the DNA double helix is separated;
- A4. and each parental strand serves as the template for the synthesis of a daughter strand;
- A5. to form two identical copies of DNA double helix;
- A6. each with one parental strand and one newly-synthesised strand;

- A7. The specific sequence of bases on one strand determines the specific order of bases on the other strand;
- A8. This is achieved by complementary base pairing;
- A9. based on Chargaff's rule;
- A10. where A pairs with T;
- A11. and G pairs with C;

- A12. DNA polymerase contains proofreading activity;
- A13. which will check/ensure the base-pairing is correct;
- A14. Any mispairing of bases will be removed;
- A15. and be replaced by the correct nucleotides;

- B1. Proofreading and repairing by DNA polymerase can lead to mistakes;
- B2. where incorrect base pairs are not excised;
- B3. There is DNA polymerase which has no proofreading activity in some living organisms;
- B4. so many mispairing of bases could occur during DNA replication to generate mutation;

- B6. Replication slippage could happen during replication;
- B7. In backward slippage;
- B8. the template shifts with respect to the new complementary strand and one region of the template strand is copied twice;
- B9. so there is an insertion mutation;
- B10. In forward slippage;
- B11. the template shifts with respect to the new complementary strand and one region of the template strand is copied twice;
- B12. so there is a deletion mutation;

(b) Discuss the importance of different events at interphase of the cell cycle in preparation for cell and nuclear division. [6]

- 1. During the G_1 phase, there is cell growth ;
- 2. and intensive cellular synthesis of mRNA and proteins e.g. histones ;
- 3. Synthesis of cellular organelles e.g mitochondria and chloroplasts;
- 4. which ensures equal division into the daughter cells;
- 5. There is synthesis of ATP;
- 6. in preparation for energy consuming processes during cell division;
- 7. During S phase, DNA replication occurs;
- 8. The amount of DNA content doubles;
- 9. due to formation of two genetically identical sister chromatids held at the centromere in each chromosome;
- 10. In the G_2 phase, there is extensive synthesis of tubulin;
- 11. to form spindle fibres which will bind to kinetochore at centromeres of chromosomes;
- 12. Centrioles replicate;
- 13. to organize spindle fibres during cell division;

(c) The Central Dogma dictates the flow of genetic information of DNA to RNA to proteins. Suggest how this statement may not be valid. [4]

- 1. RNA can assume the role of DNA as a store of genetic information;
- 2. RNA can replicate ;



3. based on complementary base pairing of nucleotides;
4. The influenza virus which makes use of replicase to form (+) strand RNA from (-) strand RNA;
5. There can be reverse flow of information from RNA to DNA;
6. HIV virus which contains RNA as genomic material;
7. and reverse transcriptase to convert RNA to DNA;
8. RNA can take on the role of proteins to catalyze reactions;
9. Ribozyme/peptidyl transferase in ribosomes;
10. catalyzes the formation of peptide bonds;
11. (4m max)

[Total: 20 marks]

QUESTION 9

(a) Explain with a named example, why cancer development is a multistep process and how it relates to changes in gene expression. [10]

1. Colorectal cancer;
2. development is a multistep process because there is an accumulation of mutations in the same cell line;
3. The first mutation is a loss-of-function mutation in one allele;
4. of APC tumour suppressor gene;
5. This results in insufficient APC protein produced;
6. The cell line partially escapes cell cycle control/ forms a polyp;
7. The second accumulated mutation is a gain-of-function mutation in one copy;
8. of Ras proto-oncogene;
9. A hyperactive Ras protein is produced;
10. This results in stimulation of cell division even in the absence of growth factor/ cell cycle proceeds without growth factor stimulation;
11. The third accumulated mutation is a loss-of-function mutation in both alleles;
12. of DCC tumour suppressor gene;
13. This results in no functional DCC protein produced;
14. Cell adhesion and cell differentiation is lost;
15. The fourth accumulated mutation is a loss-of-function mutation in both alleles;
16. of p53 tumour suppressor gene;
17. This results in no functional p53 protein produced;
18. and high rate of mutation throughout the genome occurs/ controlled cell proliferation is lost;
19. Telomerase gene activation occurs;
20. and results in an increase in telomerase activity/ production of telomerase;
21. to lengthen telomere/maintain telomere length;
22. so the cell line can continue to divide infinitely/ escape Hayflick limit/ does not enter replicative cell senescence;

(b) Describe **two** mechanisms by which oncogenes arise and the significance of these mechanisms. [6]

Any two mechanisms below:

1. Point mutation/Local DNA rearrangement;
2. can happen at proto-oncogene;
3. where there is base substitution/ base insertion/base inversion/base transposition /base deletion;
4. It produces new hyperactive protein;
5. if it happens at coding region as the nucleotide sequence is changed;
6. It could over-express the protein;
7. if it happens at the promoter/regulatory sequence;
8. The significance is transposition by transposons can lead to cancer development;; /
9. over-exposure to ionizing radiation/chemical carcinogens that create point mutation can lead to cancer development;;
10. Gene amplification;
11. as a result of replication slippage/ unequal crossing over;
12. increases in the copies of a proto-oncogene;
13. The proteins encoded will be over-produced;
14. The significance is unchecked DNA replication could lead to cancer development;;



15. Chromosomal translocation;
16. removes a portion of one chromosome;
17. and joins it to another chromosome;
18. which contain an active promoter/ actively transcribing genes;
19. This results in over-expression of the translocated gene and it becomes an oncogene;
20. Insertional mutagenesis;
21. that involves retrovirus inserting their DNA into proto-oncogene;
22. Can lead to production of a new hyperactive protein;
23. or over-expression of the protein;
24. The significance is viral infection can lead to cancer development;;

(c) Suggest why cancer is best treated at the early stage

[4]

1. Tumour can be surgically removed easily/Cancer cells can be more effectively killed by drugs;
2. Cancer formed is usually benign/not malignant;
3. Tumour is localized;
4. Metastasis has not occurred/cancer cells have not spread to distant sites;
5. Angiogenesis has not occurred/extensive blood vessels have not developed;
6. During treatment, there is less damage to other organs;
7. Few mutations at the early stage;
8. Lower chance of forming malignant/secondary tumour;
9. Less likely to have a relapse;

[Total: 20 marks]

QUESTION 1

(a) Apart from its small size, state another characteristic of recombinant insulin which makes it suitable to be produced using genetically engineered bacteria. [1]

- Human insulin is not modified after translation by the addition of sugar molecules; OR
- There is thus no need for a human expression system to ensure that post-translational modification takes place; AND
- Thus, recombinant insulin synthesized by bacteria should be active;

(b)(i) Describe how the artificial genes in the respective recombinant plasmids could be induced to undergo expression in *E. coli*. [1]

- Via the addition of lactose; to form allolactose to inactivate the *lac* repressor;

(ii) Explain how the selection of transformed bacterial cells could be carried out. [3]

- Bacterial cells successfully transformed with recombinant plasmids will be resistant to tetracycline; as they possess the tetracycline resistance gene;
- thus can survive / grow / form colonies on the tetracycline-containing agar plate; (1m max)
- These colonies transformed with recombinant plasmids will have a disrupted β -galactosidase gene / β -galactosidase gene subjected to insertional inactivation;
- which would not be transcribed and translated to give β -galactosidase enzyme;
- (substrate) X-gal would not be hydrolyzed / cleaved;
- thus the colonies remain white; (2m max)

(c) With reference to **Fig. 1.1** and **Fig. 1.2**, complete **Fig. 1.3** above by identifying the particular restriction enzyme used for the digestion of the respective plasmid DNA. [2]

	Digestion of plasmid DNA with artificial gene for A chain				Digestion of plasmid DNA with artificial gene for B chain			
Plasmid	A1	A2	A1	A2	B1	B2	B1	B2
Length of fragments (in base pairs)	406	463	1234	1235	1288	1208	448	448
	2463	2406	1635	1634	1608	1688	2448	2448
Restriction enzyme used	<i>EcoRI</i> ;		<i>PstI</i> ;		<i>PstI</i> ;		<i>EcoRI</i> ;	

(d) With reference to **Fig. 1.1**, **Fig. 1.2** and **Fig. 1.3**,

(i) state the appropriate restriction enzyme that has to be used for the digestion of the respective plasmid DNA in order to determine which has the artificial genes inserted in the correct orientation. [1]

Plasmid DNA with artificial gene for A chain – *EcoRI*;
Plasmid DNA with artificial gene for B chain – *PstI*;

(ii) explain your answers to (d)(i). [2]

- The artificial genes must be cut in such a manner that the fragments generated are of different lengths / not of the same length;
- so as to clearly distinguish between the restriction fragments generated from recombinant plasmids with the artificial gene inserted in the correct and wrong orientations;

AND

- e.g. if artificial gene A inserted in the correct orientation is cut using *PstI*, the restriction fragments generated (1235 bp and 1634 bp) are of almost the same lengths as those generated (1234 bp and 1635 bp) if artificial gene A is inserted in the wrong orientation;;

OR

- e.g. if artificial gene B inserted in the correct orientation is cut using *EcoRI*, the restriction fragments generated (448 bp and 2448 bp) are exactly of the same lengths as those generated if artificial gene B is inserted in the wrong orientation;;

(iii) determine which plasmids contain the artificial genes for the A and B chains inserted in the correct orientation. [1]

Plasmid with artificial gene for A chain inserted correctly – A2;

Plasmid with artificial gene for B chain inserted correctly – B1;

(e) Suggest **one** advantage of using human recombinant insulin produced by eukaryotic cell lines over porcine- or bovine-derived insulin. [1]

- It is identical to human insulin;
- and have no side / adverse effects / allergic / immunological reactions;

(f)(i) Explain how retroviral-mediated gene transfer can be used in gene therapy to treat diabetic patients. [2]

- Retroviruses are genetically engineered to be replication-defective;
- and carry instead the RNA version of the normal / wild-type human insulin gene;
- when the viruses enter the target cells, they release their RNA into the cytosol;
- where reverse transcriptase of the retroviruses each copies a double-stranded DNA from the RNA;
- which enters the nucleus and is integrated into the host genome;
- after which the inserted normal / wild-type human insulin gene can then be transcribed and translated / expressed to give functional insulin;

(2m max)

(ii) Suggest **one** possible reason why treatment of diabetes by gene therapy is still at the experimental stage. [1]

- Difficult to get DNA to integrate into the target cell genome;
- results can be short-lived;
- patients need to be treated on a frequent basis;
- ref to problem with controlling the activity of gene expression;
- Risk of stimulating immune response;
- virus vector may regain / develop virulence;
- incorrect insertion of gene may cause malignancy / mutagenic effects;

(1m max)

[Total: 15 marks]

QUESTION 2

(a) Describe briefly one difference between VNTRs and STRs. [1]

- The repeat unit of VNTRs is longer of 10 up to 100 nucleotide pairs in length; while STRs are repeats of shorter sequences of only 1 to 9 base pairs in length; OR
- VNTRs tend to be located at telomeric ends of chromosomes; while STRs are more evenly distributed throughout the genome;

(b) Discuss the significance of the *EcoRI* cleavage sites in this DNA fingerprinting. [2]

- The *EcoRI* cleavage sites flanking each of the three VNTR loci; are where the restriction enzyme *EcoRI* will cleave the DNA;
- to produce a series of DNA fragments of variable lengths or RFLPs; that vary in direct proportion with the number of tandem repeats; that produce a specific / unique pattern of bands; on a Southern blot.

(c) In the human population, **n** may range between 3 and 20 for VNTR locus **1**. Give the expected number of alleles of VNTR locus **1** that would occur in the human population. [1]

- 18;;

(d)(i) Write the sequence of a 10-bp probe molecule that can be used to detect all 3 VNTR loci **1** to **3**. [1]

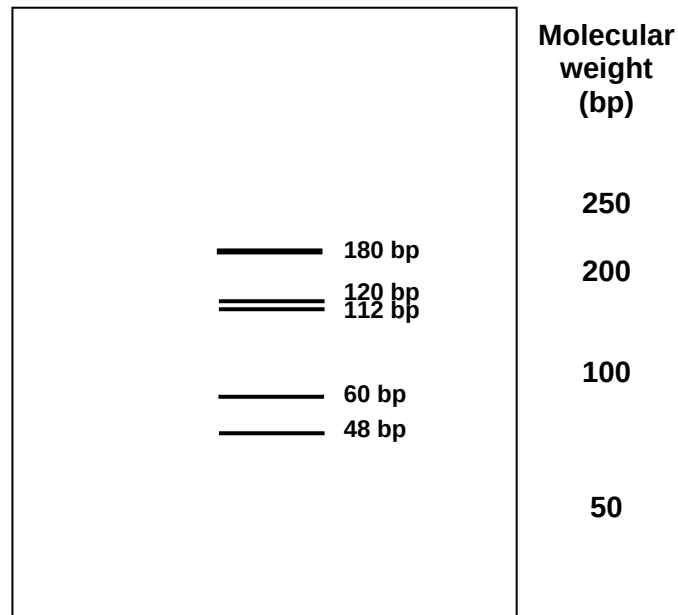
- 3'- GGCTAGCCGT – 5';; OR 5'-TGCCGATCGG-3';; (DNA probe) OR
- 3'- GGCUAGCCGU – 5';; OR 5'-UGCCGAUCGG-3';; (RNA probe)



(ii) Outline the experimental procedure, after the action of *EcoRI*, of obtaining a DNA fingerprint. [3]

- DNA restriction fragments are size-fractionated by gel electrophoresis;
- bands obtained are transferred / blotted from the gel onto a nitrocellulose or nylon membrane;
- The DNA is denatured into single strands by the alkaline buffer;
- The membrane is then incubated with radioactively / chemically labelled single-stranded DNA or RNA probes; which hybridise to complementary target sequences on the membrane;
- The membrane is washed thoroughly to remove unbound probe;
- followed by detection of the bound probe by autoradiography / chemical means; to produce a specific pattern of bands.

(e) Draw the expected results on a Southern blot below, indicating the molecular weight of each expected band. One of the expected bands is shown. [2]



- Correct positions of four bands, corresponding to 60 bp, 112 bp, 120 bp and 180 bp, with respective molecular weights indicated;;;
- with 180-kb band thicker than the rest of the bands;

(f) If three other VNTR loci with *n* values not more than 5 are used, suggest with a reason the effect on the discriminating power (ie. ability to distinguish between individuals) of RFLP analysis using these loci. [1]

- Discriminating power would be lowered / reduced;
- as with a smaller *n* value / number of alleles or allelic variants or polymorphisms at each locus, the number of allelic combinations at each locus would be smaller;

(g) Give a reason why multiple sets of primers are used in this PCR-STR analysis. [1]

- Multiple sets of primers are used, each set being complementary / binding to the particular DNA region flanking its respective target STR locus;
- for amplification of the alleles of the particular locus / multiple loci;

(h) With reference to the data in **Fig. 2.3**, deduce

(i) the alleles of the **D3S1358** locus present in the individual. [1]

- Two; allele 15s; OR homozygous; for allele 15; at the **D3S1358** locus.

(ii) the number of bands visible on the gel after gel electrophoresis. [1]

- 5;;

(i) Based on the information given above about PCR-STR and **Fig. 2.3**, explain **one** advantages of PCR-STR over the RFLP analysis of VNTR loci you described in (d)(ii). [1]

Any one of the following:



- Specimens with just trace amounts of DNA; can be analysed by PCR-STR, whereas for RFLP analysis of VNTR loci, a relatively large amount of DNA; is required.
- Even relatively degraded DNA samples can be analysed; by PCR-STR, whereas RFLP analysis requires relatively intact and non-degraded DNA;
- PCR-STR does not involve the use of restriction enzymes and is thus less expensive; unlike RFLP analysis which requires the use of costly restriction enzymes;
- PCR-STR procedures can be automated and is thus faster; compared to RFLP analysis using Southern Blotting which includes more labour-intensive; steps of blotting onto membrane, incubation with probe and probe detection.

[Total: 15 marks]

QUESTION 3

(a)(i) Identify an assumption on which this idea is based. [1]

- Ice⁻ strain will compete with Ice⁺; reducing the number of Ice⁺;

(ii) Suggest how spraying a field with the Ice⁻ strain may have an adverse effect on the crop yield. [2]

- Ice⁻ strain growing on leaves of other plants (weeds) will transfer the mutant gene / transgene; to the weeds' cells;
- Weeds would then become less susceptible to frost damage;
- Increased competition between weed and crop plants for nutrients, space for growth (mention of 1 factor);
- Crop yield will be reduced;

(b)(i) Suggest a suitable control to be used in this investigation. [1]

- Plants sprayed only with carrier solution / same volume of carrier solution;
- without bacteria Ice⁻;

(ii) Suggest the advantage in plotting such data on a log scale. [1]

- Enables a large range of numbers; to be plotted
- as bacterial growth is exponential;

(iii) From the data collected on population densities, error bars and standard deviations were calculated. Explain the link between standard deviation (SD) and the reliability of the conclusions that can be drawn from an investigation such as this. [2]

- The greater the SD, the greater the variation in the sample;
- the more unreliable is the data ;
- The greater the variation in sample size, the less likely it is that any differences are significant / cannot conclude that a difference exists between samples;
- more likely to be due to chance;

(c) With reference to **Fig. 3.1**, explain whether or not the results of this investigation indicate that the Ice⁻ strain would be

(i) likely to spread to plants other than the crop plants on which they were sprayed [1]

- Not likely to spread with its short-term effects;
- survival only about 6 weeks / from June to mid July;

OR

- Peak immediately after 1-2 weeks of spraying from 10³ cells / gram fresh mass of plant to more than 10⁶ cells / gram of plant fresh mass;
- relatively short survival time;

(ii) effective in reducing ice damage to the crop on which they were sprayed. [2]

- Only sprayed in June/spring, not sprayed in winter/cold weather;
- Damage only when there is frost/cold conditions/winter;
- Whether bacteria grow in winter/rate of growth may be slow in winter is unknown;

(1m max)



- Large number of Ice⁻ bacteria used;
 - so could be effective;
 - in outcompeting other bacteria for growth;
 - Overall low relative population of Ice⁻ so may not be effective;
- (1m max)

(d) Explain how the addition of nutrients may affect plasmid transfer. [2]

- Number of plasmid transfers increase from 0.2 per 10⁷ cells to 0.5 per 10⁷ cells;
- Nutrients are respiratory substrates; that can be metabolized to produce energy in the form of ATP;
- Ensures that bacteria are likely to be metabolically active;
- Larger number of bacteria will result from addition of nutrients;
- Increases likelihood of contact between recipients and donors; via formation of sex pili;

(e) Suggest why sterilizing the soil before adding *B. cereum* and *B. subtilis* may lead to a greater frequency of plasmid transfer between these two species. [1]

Any one of the following:

- Removes competitors; so greater number of *Bacillus* species;
- Any contact between species will be between these two; and no bacterial strains;

(f) Explain why the human intestine is a particularly important site of transfer. [2]

- In human intestine, favourable / optimal conditions; produced by nutrients / from digested food;
 - Warm conditions; favour rapid multiplication;
 - Large population of bacteria population ingested; so greater probability of transfer;
- (1m max)

AND

- Antibiotics taken orally;
 - will favour in selection of antibiotic-resistant strains of bacteria;
 - antibiotic-resistant strains will thrive in intestines;
 - As such, antibiotic treatment may not be successful in humans;
- (1m max)

[Total: 15 marks]

Paper 3 (APPLICATIONS) Section B: Free Response

QUESTION 4

(a) Genetic engineering applied to disease therapies can harness the unique properties of stem cells. Explain the basis of these unique properties of stem cells. [8]

- (1) The basis for the unique properties of stem cells are: mitosis; and
- (2) internal; and external signals;

Mitosis

- (3) Stem cell are capable of dividing; and
- (4) replenishing their own population;
- (5) to produce genetically identical / clonogenic cells;
- (6) with the capability for long-term self-renewal;
- (7) These cells undergo mitosis, preserving a population of undifferentiated cells;
- (8) that are genetically identical to each other and to the parent cell; while
- (9) steadily producing a stream of differentiating cells;.
- (10)that may differ in size, shape, and/or composition, or their genes may be in different states of activity or potential activity;.

Internal signals

- (11)During stem cell differentiation;
- (12)internal signals may activate certain genes;
- (13)or inactivate other genes in an intricately regulated fashion;.
- (14)Consequently, a differentiated cell develops specific structures;
- (15)and performs certain functions;.
- (16)Internal signals may confer different fates; on the two cells if they are exposed to different signals.

External signals



- (17) External signals for cell differentiation include chemicals secreted by other cells;
- (18) physical contact with neighboring cells;
- (19) and certain molecules in the microenvironment;
- (20) The ability of stem cells both to self-renew; and to differentiate into different cell types is tightly regulated by the cells and molecules in this environment;
- (21) The external signal may also act through intercellular signals;.

(b)(i) Describe the procedure for producing transgenic salmon.

[7]

- (1) Scientists duplicate the DNA carrying the genetic information for the salmon growth hormone / construct the cDNA of salmon growth hormone;.
- (2) Recombinant DNA plasmids are subsequently constructed;.
- (3) containing an antifreeze protein gene promoter from ocean pout ligated to;
- (4) the cDNA for salmon growth hormone;.
- (5) Subsequently, the recombinant plasmids are introduced into bacterial host cells using the heat shock treatment / electroporation;.
- (6) When the bacteria are allowed to grow *in vitro*, they produce billions of bacteria carrying copies of the plasmid carrying the growth hormone gene;.
- (7) Plasmids carrying the growth hormone gene are then isolated from the bacteria;.
- (8) and genetically edited, changing its circular structure into linear bit of DNA / gene cassettes;
- (9) which are subsequently directly injected into / mixed with fertilized fish eggs;
- (10) so that the eggs absorb the DNA;.
- (11) The eggs are allowed to hatch;.
- (12) producing a school of transgenic fish;.
- (13) that are identified by their larger size;
- (14) and are used to create a breeding stock of the new, faster-growing variety;.

(ii) Explain how farming of transgenic salmon in open seas could pose a potential environmental hazard.

[5]

- (1) The escape of transgenic salmon; from salmon farms into the open sea
 - (2) may permit the release of their added growth hormone genes;
 - (3) through interbreeding with wild populations;.
 - (4) The resulting hybrids may show reduced survival; compared with wild salmon.
 - (5) Transgenic salmon mature earlier;
 - (6) and hence, are larger and appear more attractive to females;
 - (7) consequently outpacing wild salmon in the mating game;
 - (8) The dominance of transgenic salmon in the reproductive process could lead to the eventual extinction of the species;.
 - (9) Transgenic salmon grow much faster;
 - (10) and are more aggressive;.
 - (11) They may displace wild salmon out of their native habitats;
 - (12) by competing with wild salmon for resources;
 - (13) hence upsetting the ecological balance of the environment;.
- (5m max)

[Total: 20 marks]